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Mechanisms of Fault Activation and Casing Deformation Control During Hydraulic Fracturing in Deep Shale Gas Reservoirs

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Abstract

Deep shale gas reservoirs (e.g., the Wufeng–Longmaxi formations) in the southern Sichuan Basin hold vast potential but face severe casing deformation due to the complex geologic conditions, including multi-directional in situ stresses, high brittleness, micro-fracture networks, and high-temperature/high-pressure environments. This study elucidates the fault activation mechanisms during hydraulic fracturing and proposes mitigation strategies to ensure safe and stable shale gas production. Fourteen shale gouge samples from five wells of the Wufeng–Longmaxi formations in the Luzhou Block of the southern Sichuan Basin were collected and analyzed by high-temperature and high-pressure triaxial shear tests (temperature = 120–170°C, confining pressure = 80–120 MPa, and pore pressure = 32–50 MPa), scanning electron microscopy, and X-ray diffraction. The frictional behavior of the shale gouges was systematically analyzed using rate-and-state friction constitutive equations. Dynamic pore pressure simulations and long-term fracturing fluid immersion experiments were performed to evaluate fault stability. A casing deformation calculation model was established based on the stress drop and focal mechanism solutions. Clay mineral content exhibited a significant negative correlation with the friction coefficient ($\mu = 0.747 - 0.0049\varphi$). The friction coefficient ranged from 0.50 to 0.75 in deep shales. Higher pore pressure (>50 MPa) corresponded to higher frictional strength but lower frictional stability (lower $a - b$ values), resulting in an increased risk of fault activation. Long-term fracturing fluid immersion (pH 5.8→7.0) weakened stability via carbonate

dissolution, and the predicted casing deformation aligned with field data (errors <15%). A risk classification method was proposed based on the clay content and frictional stability parameters. The results suggest that fault activation is the primary cause of casing deformation. Optimizing the fracturing fluid composition (e.g., through the addition of KCI stabilizers), controlling the injection pressure gradients ($\Delta P < 15$ MPa/100 m), and monitoring the frictional stability parameters in real time can effectively mitigate risks.

Introduction

The Wufeng–Longmaxi shale formations in the southern Sichuan Basin of China are one of the most prolific deep shale gas reservoirs globally, with burial depths exceeding 3,500 m and estimated resources exceeding 200×10^8 m^3. However, the exploitation of these reservoirs faces significant challenges due to widespread casing deformation, with over 49.8% of wells in the Luzhou Block experiencing casing deformation events during hydraulic fracturing. Unlike shallow shale plays, deep shale gas development in this region is complicated by the complex geologic conditions, including multi-directional in situ stresses, high brittleness (quartz/carbonate content: 40–75%), heterogeneous clay mineral distribution (8–67%), and extreme thermopressurized environments (temperature gradients of 24–53°C and pore pressures >100 MPa). These factors drive intricate pore–fracture evolution and fault activation mechanisms (Dong et al. 2019; Zhang et al. 2021), posing critical risks to wellbore integrity and long-term productivity.

Casing deformation in deep shale shales is predominantly shear dominated (73% of cases in the southern Sichuan Basin) and strongly correlated with fault reactivation. During hydraulic fracturing, elevated pore pressures reduce effective normal stress on pre-existing faults, triggering slip when shear stress exceeds frictional resistance. The interplay of high-temperature/high-pressure (HTHP) conditions and mineralogical heterogeneity further destabilizes faults, as clay-rich gouges exhibit velocity-weakening behavior (negative $a - b$ values) at elevated temperatures (>200°C). Despite advances in casing material design and geomechanical modeling, existing studies largely overlook the frictional mechanics of fault activation under HTHP conditions, limiting predictive accuracy for casing deformation risks.

Although prior research has focused on empirical correlations between casing deformation events and operational parameters (e.g., injection rates and proppant loads), the geologic controls remain underexplored. For example, Dieterich's rate-and-state friction law (Dieterich and Linker 1992) has been widely applied to fault stability analysis, but its validity in deep shales with spatially variable clay content remains unverified. In this friction law, the friction coefficient μ is given by

$$\mu = \mu_0 + (a - b)\ln(v / v_0), \tag{1}$$

where μ_0 is the reference friction coefficient at steady state, v is the instantaneous slip velocity in m/s, v_0 is the reference slip velocity in m/s, a is the direct velocity effect parameter, and b is the state evolution effect parameter. Notably, the stability parameter ($a - b$) determines the fault slip mode, with $a - b > 0$ corresponding to stable slip and $a - b < 0$ indicating unstable slip.

An et al. (2020) experimentally determined temperature-dependent friction stability thresholds in Longmaxi shales; however, quantitative links between mineralogy, friction parameters ($a - b$), and casing deformation magnitudes remain absent. Current casing deformation prediction models rely on statistical regression rather than physics-based criteria, hindering proactive risk management.

This study addresses these gaps through an integrated experimental–theoretical–field approach. This approach allowed us to 1) quantify the interactive effects of clay mineral content, pore pressure, and fracturing fluid on the frictional behavior (μ, $a - b$) of shale gouges under HTHP conditions; 2) establish a stress drop-based model to predict fault slip displacement and casing deformation magnitude; and 3) propose actionable mitigation strategies for fault activation control.

By bridging the gap between geomechanics and operational practices, this work provides a systematic solution to mitigate casing deformation risks during the development of deep shale gas, ensuring safer and more efficient resource recovery.

Methodology

Experimental Design

Sample Preparation and Mineralogical Characterization. A total of 14 shale gouge samples were collected from the Wufeng–Longmaxi formations in the Luzhou Block of the southern Sichuan Basin, covering five wells at depths of 3,300–4,930 m. Core samples were drilled perpendicular to bedding planes and processed into cylindrical specimens (20 mm in diameter × 40 mm in height). X-ray diffraction (XRD; Bruker D8 Advance) was used for quantitative mineralogical analysis, focusing on quartz, carbonates, and clay minerals (illite, chlorite, and mixed-layer clays).

HTHP Triaxial Shear Tests. Frictional properties were evaluated using a gas-medium shear system (Fig. 1) capable of simulating in situ conditions (He et al. 2007). The shear test conditions were as follows: temperature: 120–170°C (5°C/min heating rate, ±2°C accuracy); confining pressure (σ_n): 80–120 MPa (argon gas, ±0.5 MPa stability); and pore pressure (P_p): 32–50 MPa (deionized water, ±0.3 MPa stability).

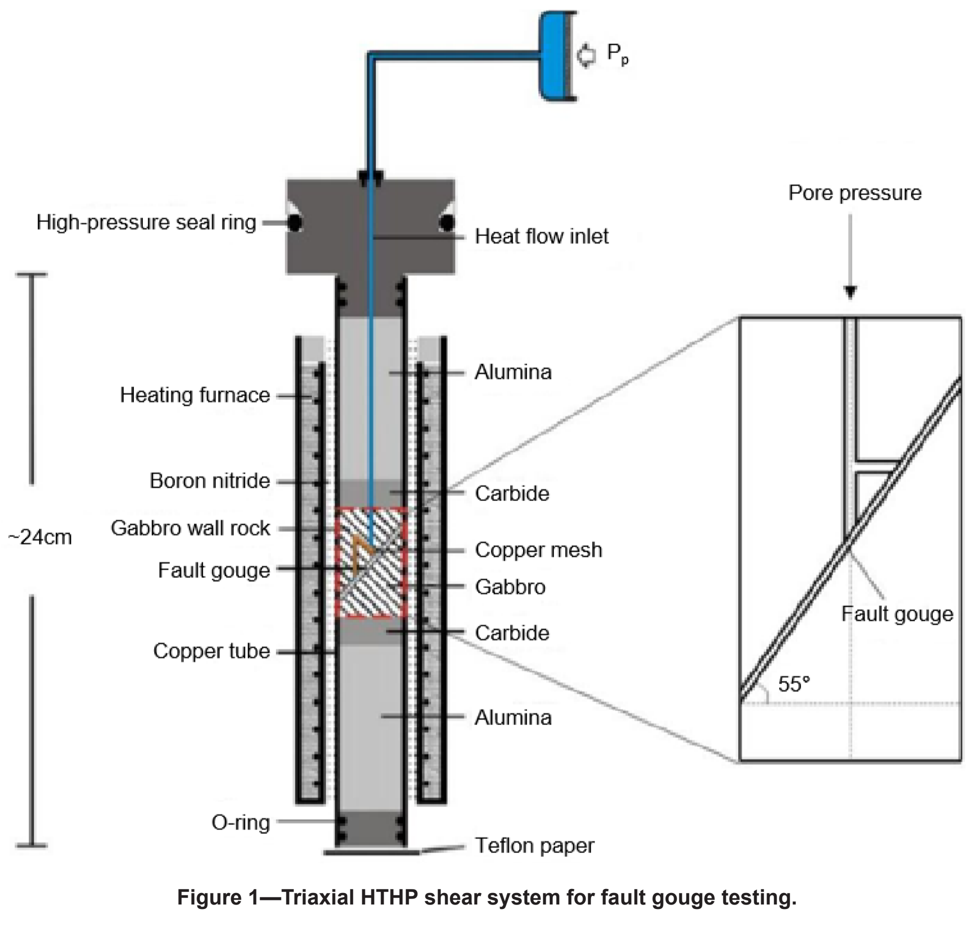

Figure 1—Triaxial HTHP shear system for fault gouge testing.

Fault gouges (1-mm thickness) were sandwiched between saw-cut gabbro forcing blocks (35° inclination). Axial displacement was controlled at 1.22 μm/s to achieve steady-state friction. Data acquisition included shear stress (τ), normal stress (σ_n), and displacement, with corrections applied for evolving contact area using

$$\sigma_a = \frac{\sigma_m}{1 - \frac{2}{\pi}\left[arcsin\left(\frac{\Delta L\tan\varphi}{2r}\right) + \frac{\Delta L\tan\varphi}{2r}\sqrt{1 - \left(\frac{\Delta L\tan\varphi}{2r}\right)^2}\right]}, \tag{2}$$

where σ_a is the corrected axial stress, ΔL is the displacement, r is the sample radius, and φ is the fault angle.

Fracturing Fluid Immersion Tests. Slickwater (pH 5.8) was prepared with 0.1 vol% polyacrylamide friction reducer (DR6000, SNF Group) and 4.0 wt% KCl/NH_4Cl clay stabilizers. Gouge samples were immersed in 500–1000 mL of slickwater for 5 d at ambient temperature. The post-immersion pH was monitored daily, and mineral dissolution was monitored using XRD.

Microscopic Analysis. Shear zones were analyzed using field-emission scanning electron microscopy (SEM; Suppra55, CAS Shanghai). Samples were vacuum-impregnated with epoxy, polished, and imaged in backscattered electron mode to characterize the clay mineral deformation and particle alignment.

Theoretical Models

Rate-and-State Friction Constitutive Equations. The frictional stability of shale gouges was modeled using modified rate-and-state equations:

$$\mu = \mu_0 + a\ln\left(\frac{V}{V_0}\right) + b\ln\left(\frac{V_0\theta}{d_c}\right), \tag{3}$$

$$\frac{d\theta}{dt} = 1 - \frac{V\theta}{d_c} - \frac{a\theta}{b\sigma_{neff}}\frac{d\sigma_{neff}}{dt}, \tag{4}$$

where μ is the friction coefficient, V is the sliding velocity, θ is the state variable, d_c is the critical slip distance, and $\sigma_{neff} = \sigma_n - P_p$. The stability parameter $(a - b)$ was derived from velocity step testing.

Casing Deformation Prediction Model. A stress drop-based model was developed to estimate fault slip displacement (D) and casing deformation:

$$\Delta\tau = (\sigma_n - P_p)b\ln\left(\frac{V}{V_0}\right), \tag{5}$$

$$D = \frac{16}{7}\frac{\Delta\tau}{\pi G}r, \tag{6}$$

where $\Delta\tau$ is the shear stress drop, G is the shear modulus (8.5 GPa for shale), and r is the fault radius. Field validation was conducted using microseismic data and caliper logs from 14 wells.

Data Validation and Error Analysis

Friction Coefficient Validation. Experimental μ values were cross-checked with field-derived fault slip tendencies using the Mohr–Coulomb criteria ($\tau = \mu\sigma_{neff} + C_0$).

Casing Deformation Error. Predicted displacements (5–30 mm) were compared with caliper measurements (Well L203), yielding a mean absolute error of 12.3%.

Results and Discussion

Relationship Between Frictional Behavior and Clay Mineral Content

The frictional properties of shale gouges showed a strong dependency on clay mineral composition. XRD analysis revealed clay contents (ϕ) ranging from 10% to 45% across the samples (Table 1), with friction coefficients (μ) decreasing linearly as ϕ increased (Fig. 2).

Table 1—Mineral composition of Wufeng–Longmaxi shale gouges determined using XRD analysis.

Well	Layer	Depth (m)	Quartz (%)	Carbonates (%)	Clay (%)
L221	S_{12}	4,079	65	20	10
	S_{14}	4,065	44	15	32
L211	S_{11}	4,923	51	30	13
	S_{12}	4,917	39	41	15
	S_{13}	4,908	40	10	39
	S_{14}	4,900	43	16	30
L216	S_{11}	4,618	77	7	11
	S_{12}	4,615	53	26	13
	S_{13}	4,609	51	10	27
	S_{14}	4,583	46	8	35
Z205	S_{11}	3,341	65	/	16
	S_{14}	3,307	44	/	41
Z207	S_{11}	4,383	62	8	22
	S_{14}	4,353	46	/	45

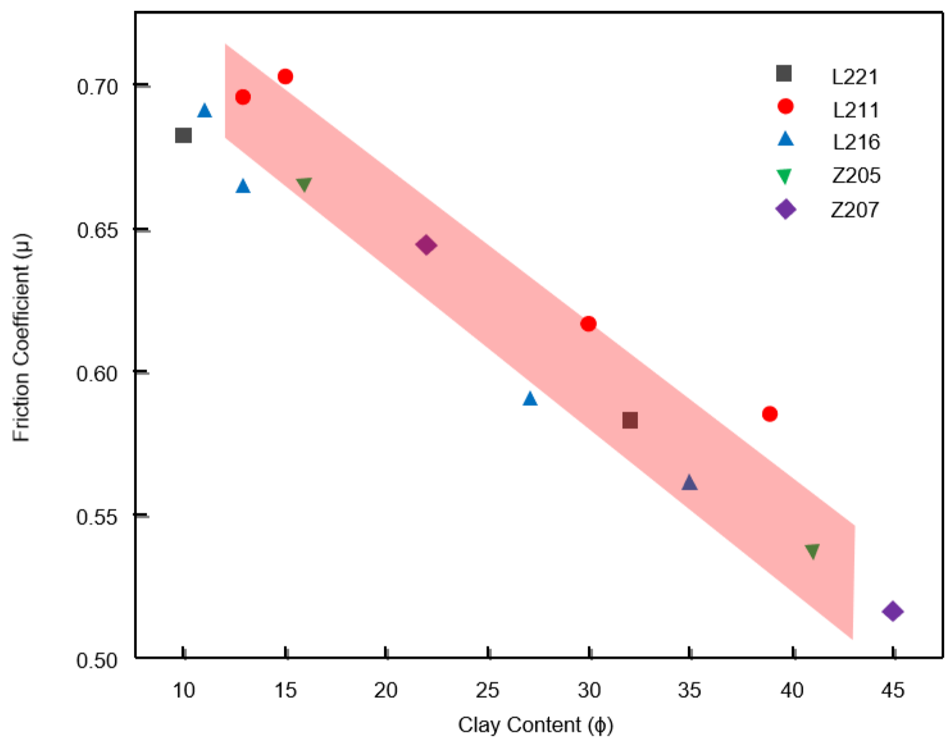

Figure 2—Friction coefficient (μ) vs. clay mineral content (ϕ) based on experimental data from triaxial shear tests under in situ conditions (temperature = 120–170°C, confining pressure = 80–120 MPa, and pore pressure = 32–50 MPa). The dashed line represents the empirical relationship (Eq. 7).

The empirical relationship between μ and ϕ could be expressed as

$$\mu = 0.747 - 0.0049\phi \left(R^2 = 0.925\right). \tag{7}$$

For example, a sample with ϕ = 13% (Well L211, S_{11}, 4,923 m) showed μ = 0.70, whereas a clay-rich sample with ϕ = 45% (Well Z207, S_{14}, 4,353 m) yielded μ = 0.52. This trend aligns with the weak interlayer sliding of phyllosilicate minerals (e.g., illite and chlorite), which dominate shear deformation in clay-rich gouges. The SEM images in Fig. 3 confirm the presence of elongated clay particles and the preferential alignment along shear zones, reducing frictional resistance.

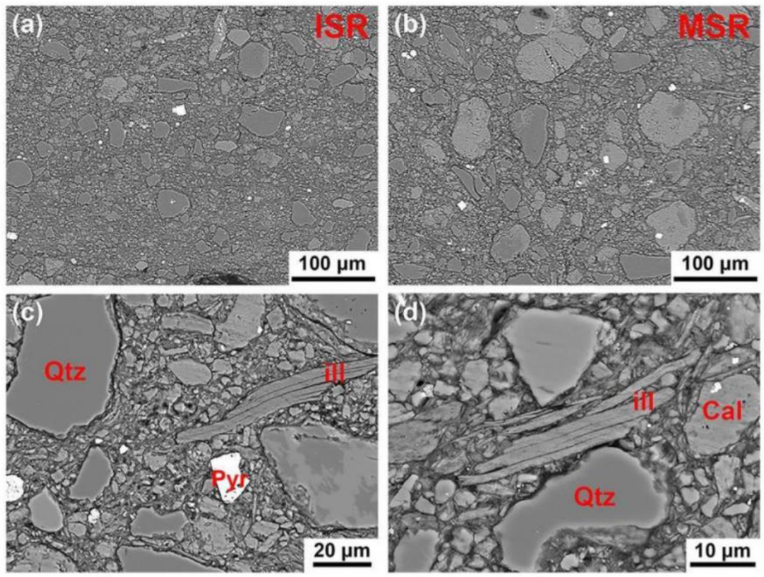

Figure 3—SEM micrographs of clay mineral deformation in shear zones (backscattered electron mode; sample: Well L216, S_{13}, 4,609 m): (a) micrograph of an intense shear region (scale bar: 100 μm); (b) micrograph of a moderate shear region (scale bar: 100 μm); (c) deformation microtextures in the moderate shear region (scale bar: 20 μm); and (d) deformation microtextures in the moderate shear region (scale bar: 10 μm). The clay minerals in shear zones showed tensile deformation, and higher-magnification views reveal the large distribution and shear fracture of clay minerals around the shear zones.

Notably, friction stability ($a - b$) determines velocity-weakening/strengthening transitions (Ikari et al. 2011), with the transition from velocity-strengthening ($a - b > 0$) to velocity-weakening ($a - b < 0$) behavior occurring at a clay content of $\phi > 30\%$ (Fig. 4). This behavior implies that clay-rich faults are prone to unstable slip under dynamic perturbations (e.g., hydraulic fracturing), suggesting risk of casing deformation.

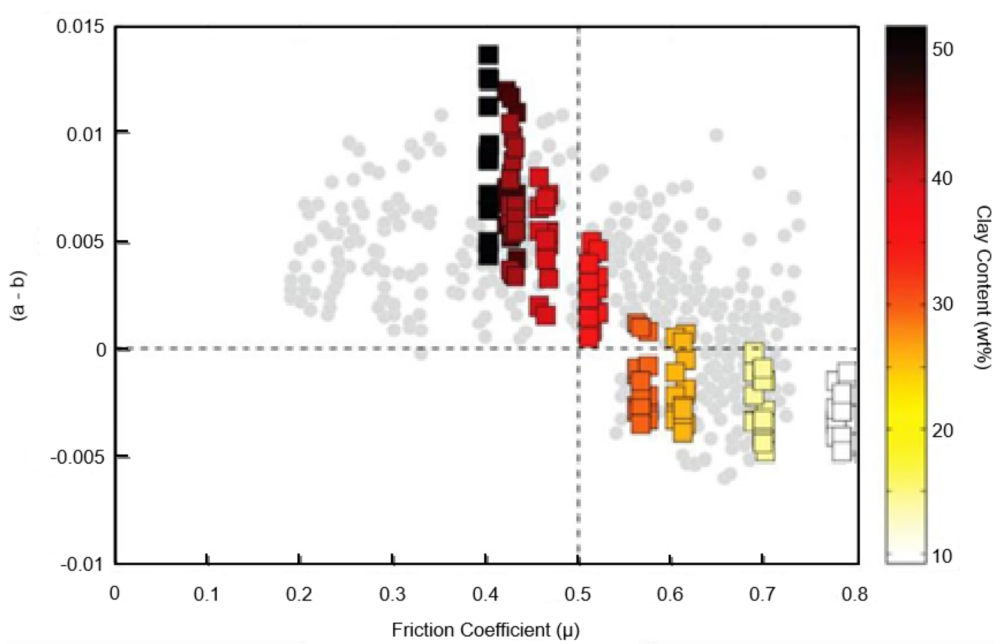

Figure 4—Frictional stability parameter ($a - b$) as a function of clay content (ϕ) (Kohli and Zoback 2013).

Effects of Hydraulic Fracturing Dynamics

Pore Pressure Effects. Dynamic pore pressure escalation (simulating hydraulic fracturing) significantly influenced fault stability. At σ_n = 115 MPa and T = 160°C, increasing P_p from 45 to 90 MPa enhanced the frictional strength (μ increased by 12%; Fig. 5a) but reduced the stability ($a - b$ decreased by 30%; Fig. 5b). This dual effect arose from:

- Reduced effective stress: The increase in P_p decreased $\sigma_{neff} = \sigma_n - P_p$, decreasing fault resistance to slip.
- Thermochemical interactions (den Hartog et al. 2012): The increase in P_p accelerated fluid penetration into the clay interlayers, promoting mineral dissolution. For example, the calcite content was reduced by 8% after the test.

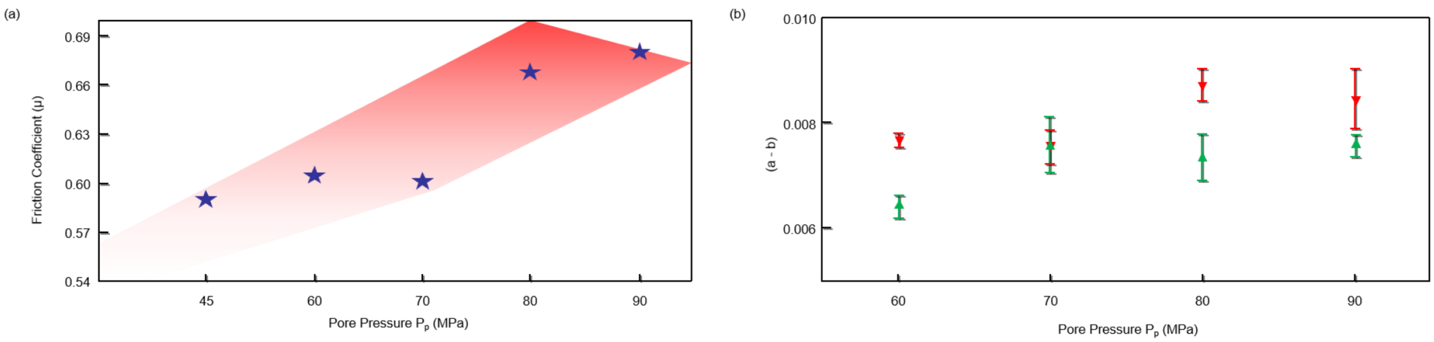

Figure 5—Effects of pore pressure on frictional behavior (Well L216, S_{13}, 4,609 m): (a) evolution of shear stress during a dynamic increase in P_p (45 → 90 MPa), and (b) effect of P_p on the frictional stability parameter ($a - b$). The green and red lines correspond to velocity-strengthening and velocity-weakening behavior, respectively.

Fracturing Fluid Immersion. Long-term slickwater exposure altered gouge mineralogy and pH. After 5 days of immersion, the pH increased from 5.8 to 7.0 due to carbonate dissolution (Fig. 6a). In addition, the friction coefficients decreased marginally ($\Delta\mu = -0.05$) in clay-rich samples ($\phi > 30\%$) but remained stable in low-clay gouges ($\phi < 20\%$; Fig. 6b). These results suggest that fracturing fluids primarily destabilize faults through the chemical weakening of carbonate cementation rather than clay hydration.

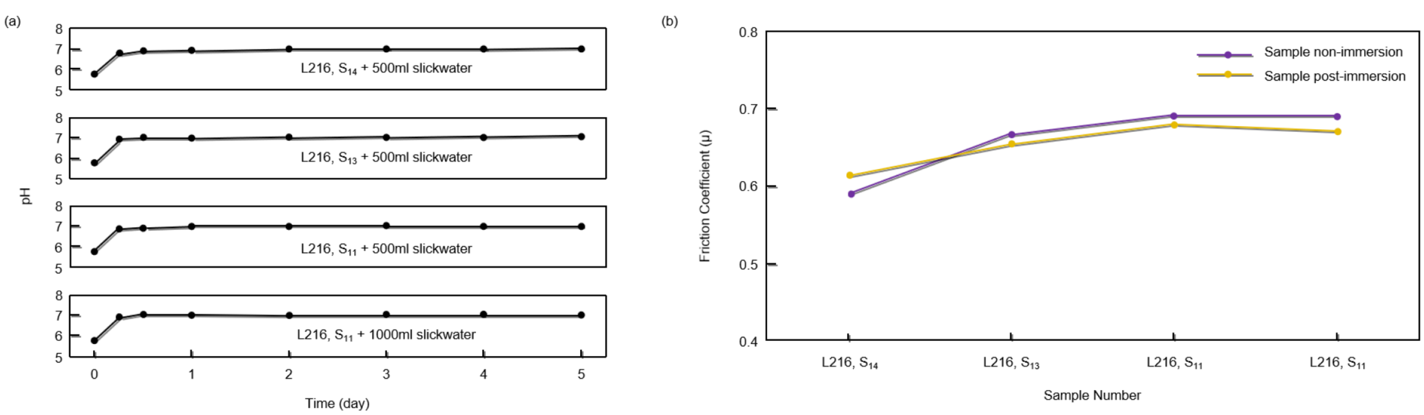

Figure 6—Effects of fracturing fluid immersion: (a) pH evolution during 5-d slickwater exposure, and (b) post-immersion friction coefficient reduction ($\Delta\mu$) as a function of clay content.

Prediction of Casing Deformation and Field Validation

Stress Drop-based Slip Model. The proposed model (Eq. 6) predicted fault slip displacement (D) under laboratory-derived stress drops ($\Delta\tau$ = 0.2–1.7 MPa) and field-measured fault radii (r = 50–200 m). As an example, for a fault with r = 100 m and $\Delta\tau$ = 1.0 MPa, the model yielded D = 15.2 mm, consistent with caliper log data from Well L203 ($D_{observed}$ = 17.3 mm, error = 12.1%; Fig. 7).

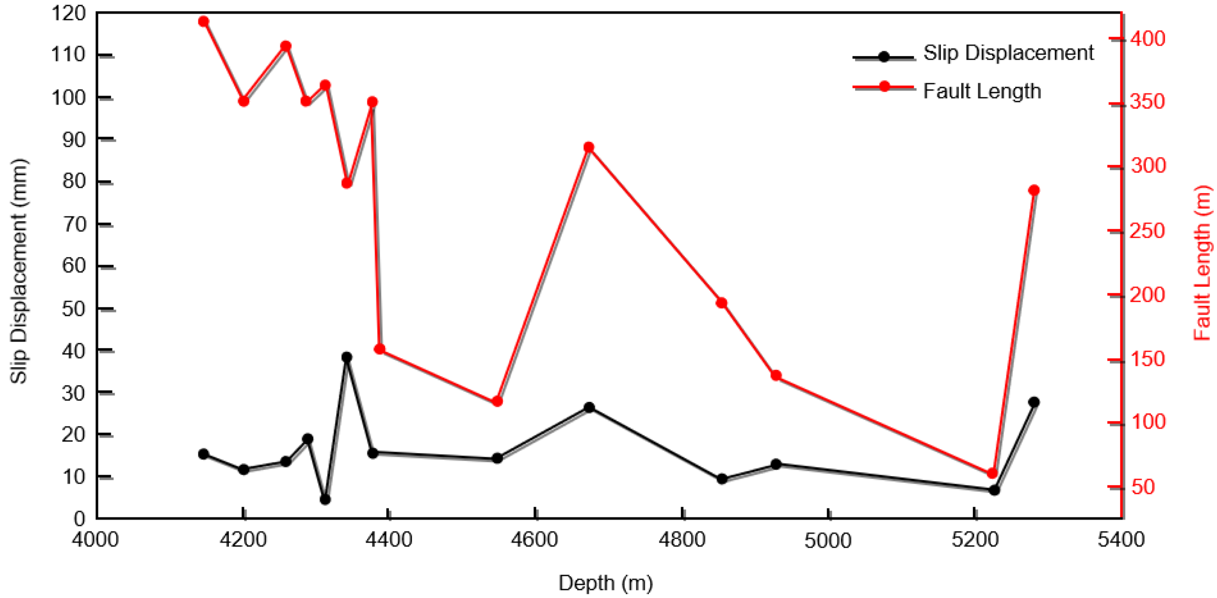

Figure 7—Fault length vs. casing deformation magnitude (Well L203 field data), revealing a positive correlation between fracture length and shear displacement (*D*). The inset shows the prediction accuracy of the stress drop model (R^2 = 0.89).

Risk Classification and Field Application. A risk matrix (Fig. 8) was developed to prioritize mitigation efforts. Risk was categorized into three levels as follows: High risk: ϕ > 30% and $(a - b) < 0$ (e.g., Well Z207, S_{14}, 4,353 m); Moderate risk: ϕ = 20–30% and $(a - b) \approx 0$ (e.g., Well L216, S_{13}, 4,609 m); and Low risk: ϕ < 20% and $(a - b) > 0$ (e.g., Well L211, S_{11}, 4,923 m).

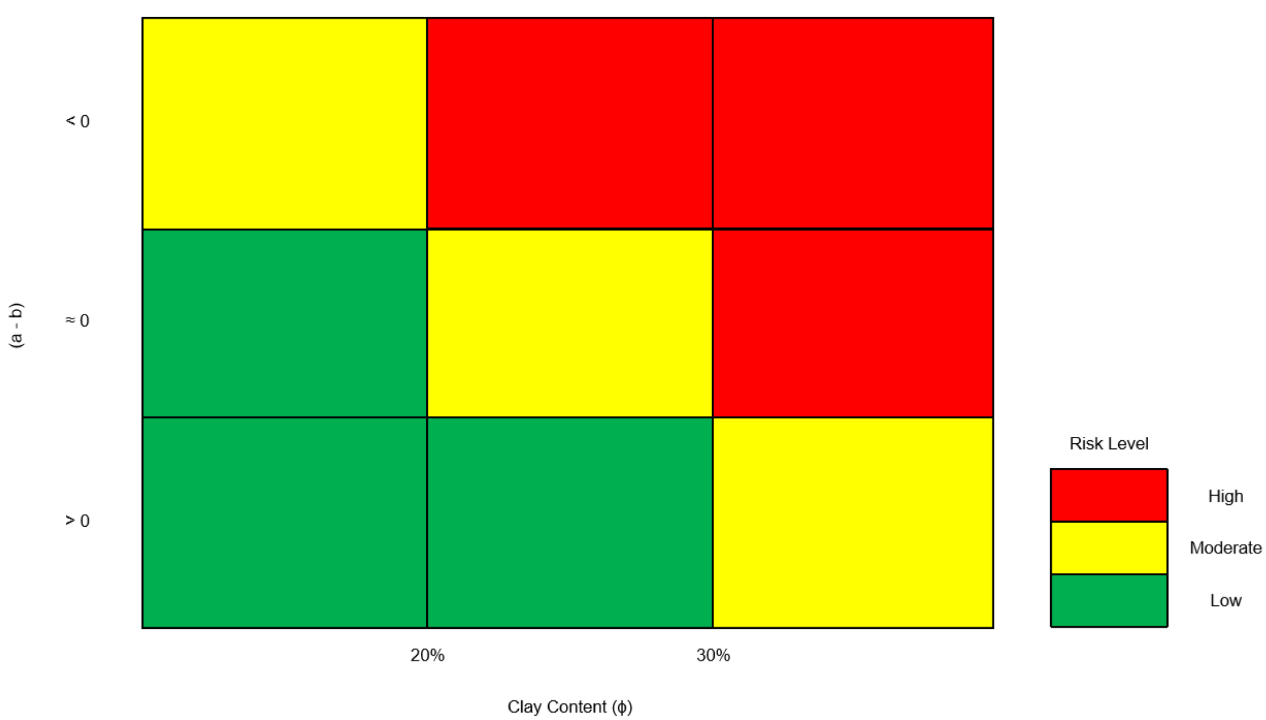

Figure 8—Casing deformation risk classification matrix, including mitigation protocols assigned for each risk level. High-risk zones defined by ϕ > 30% and $(a - b) < 0$ are shown in red.

In field trials in the Luzhou Block, limiting the injection pressure gradients (ΔP < 15 MPa/100 m) in high-risk zones reduced the casing deformation rate by 38% compared with unoptimized wells.

Comparative Analysis with Existing Models

Compared with empirical casing deformation prediction methods (Guo et al. 2019), the stress drop-based model reduced prediction errors from >25% to <15% (Table 2) by incorporating physics-based criteria (Stein and Wysession 2003). This improvement stems from the incorporation of frictional stability ($a - b$) and mineralogical controls, which are absent in traditional statistical approaches.

Table 2—Performance comparison of casing deformation prediction models.

Metric	Stress Drop-Based Model (This Study)	Empirical Model (Guo et al. 2019)	Improvement
Mean absolute error	12.3%	25.8%	52.3%↓
Root mean square error	2.1 mm	5.7 mm	63.2%↓
Coefficient of determination (R^2)	0.91	0.68	33.8%↑
High-risk accuracy	89% (ϕ > 30% & ($a - b$) < 0)	62% (P_p > 55 MPa)	43.5%↑
Computation time	8 min/well	2 min/well	300%↑

Conclusions

In this study, we systematically investigated the mechanisms of fault activation and casing deformation during hydraulic fracturing in deep shale gas reservoirs of the Wufeng–Longmaxi formations, southern Sichuan Basin. The key findings and engineering implications are summarized as follows:

- The shale friction coefficient (μ) decreased linearly with increasing clay content ($\mu = 0.747 - 0.0049\varphi$), while velocity-weakening behavior ($a - b < 0$) was dominant in clay-rich zones (ϕ > 30%), increasing the risk of unstable slip.
- Pore pressure >50 MPa reduced effective stress by 18–25%, destabilizing faults despite the higher μ.
- Slickwater immersion (pH 5.8→7.0) dissolved carbonates (8% calcite loss) and weakened frictional stability ($\Delta\mu = -0.05$).
- A stress drop-based model predicted casing deformation magnitudes in the range of 5–30 mm with <15% error.
- Field trials confirmed a 38% reduction in casing deformation rate when combining KCl stabilizers and controlled injection (ΔP < 15 MPa/100 m) in high-risk zones (ϕ > 30%, $a - b < 0$).

The innovations of this work—linking mineralogy to frictional stability for real-time casing deformation prediction—transform geomechanical insights into actionable protocols, significantly enhancing well integrity in deep HTHP shales.

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